Mathematics for Computing (IT1030)

2020 – Semester 1

Lab Sheet – 02

Relational expressions

• Expressions that are true or false (*Boolean* or logical expressions)

> greater than

< less than

>= greater than or equals

<= less than or equals

== equality

~= inequality

It will display "1" if a logical expression is true and "0" otherwise.

1. Consider the following expressions.

a) $2 < 3$

b) $3 > 5$

c) $5 >= sqrt(10\text{^}2/4)$

d) $3 <= 5$

$a = 3;$

$b = 27\text{^}(1/3);$

$a <= b$

f) $2\sim = 4;$

e)

Logical operators

The truth of the Boolean expression is computed by combining the truth values of the corresponding elements of the component expressions.

|| - OR
&& - AND
~ - NOT

2. Consider the following expressions.
   a) $2 < 3$ AND $4 > 7$
   b) $\sim(2 < 3)$ OR $4 == 3$

   c) $P = 2 > 3$;
      $q = 4 < 5$;

      a. P or q
      b. P and q
      c. ~p
      d. ~q
      e. (~p or q) and q
      f. (P and q) or (~q or p)
      g. P or ~q and (p or q)

String Conversions

Octave supports various kinds of conversions between strings and numbers. As an example, it is possible to convert a string containing a hexadecimal number to a floating point number.

Ex:- hex2dec("D2")
Ans:- 210

3. Consider the followings.

a) Convert the following Hexadecimal numbers to Decimal.
   a. D4

b. C41
c. 2F2C

b) Convert the following Decimal numbers to Hexadecimal.
   a. 12
   b. 1234
   c. 534

c) Covert the following Decimal numbers to Binary.
   a. 123
   b. 245
   c. 771

d) Convert the following Binary numbers to Decimal
   a. 110010
   b. 1010
   c. 111000111

e) Convert the following Decimal numbers to Octal, Hexadecimal and Binary using **dec2base**

   a. 123
   b. 54
   c. 223

Exercise

A nuclear power station has a safety system based on three inputs to a logic circuit (network). A warning signal (S = 1) is produced when certain conditions in the nuclear power station occur based on these three inputs.

Inputs	Binary values	Description of plant status
T	1	Temperature > 115°C
	0	Temperature <= 115°C
P	1	Reactor pressure > 15 bar
	0	Reactor pressure <= 15 bar
W	1	Cooling water > 120 litres/hour
	0	Cooling water <= 120 litres/ hour

A warning signal (S = 1) will be produced when any of the following occurs:

(a) Temperature > 115ºC and Cooling water <= 120 litres/ hour

or

(b) Temperature <= 115ºC and Reactor pressure > 15 bar

or

(c) Cooling water <= 120 litres/hour

Write a function called “Signal” using Octave to input Temperature, Reactor pressure and cooling water and display “Alarm On” if warning signal is being produced and display “Alarm off” otherwise.